


a. An African cycad




b. *Ginkgo biloba,* a native of China

**Figure 23.19 Three groups of gymnosperms.** **a.** Cycads may resemble ferns or palms, but they are cone-producing gymnosperms. **b.** A ginkgo tree has broad leaves and fleshy seeds borne at the end of stalklike megasporophylls. **c.** *Ephedra,* a type of gnetophyte, is a branched shrub. This specimen produces pollen in microsporangia. **d.** *Welwitschia,* another type of gnetophyte, produces two straplike leaves that fray and split. (a): (cycad): Hoberman Collection/Alamy Stock Photo; (cycad cone): PlazaCameraman/iStock/Getty Images; (b): (Ginkgo leaves): blickwinkel/Koenig/Alamy Stock Photo; (ovule): Ken Robertson, University of Illinois/INHS; (c): (plant): Evelyn Jo Johnson/McGraw Hill; (microsporangia): Karl J. Niklas; (d): TravelMuse/Alamy Stock Photo




c. *Ephedra*, a type of gnetophyte




d. *Welwitschia*, a type of gnetophyte

40 kg (Fig. 23.19*a*). Cycads exhibit the life cycle of a gymnosperm, except they are pollinated by insects rather than by wind. Also, the pollen tube bursts in the vicinity of the archegonium, and multiflagellated sperm swim to reach an egg.

Cycads were plentiful in the Mesozoic era at the time of the dinosaurs, and it's likely that dinosaurs fed on cycad seeds. Now, cycads are in danger of extinction, because they grow very slowly.

### *Ginkgoes*

Although **ginkgoes** are plentiful in the fossil record, they are represented today by only one surviving species of tree, the *Ginkgo biloba*. Ginkgoes are **dioecious,** which means that a single plant produces either male or female reproductive structures, but not both (Fig. 23.19*b*). The fleshy seeds, which ripen in the fall, give off such a foul odor that male trees are usually preferred for planting. Ginkgo trees are resistant to pollution and do well along city streets and in city parks. Ginkgo is native to China, and in Asia, ginkgo seeds are considered a delicacy. Extracts from ginkgo trees have been used to improve blood circulation.

Like cycads, the pollen tube of ginkgo bursts to release multiflagellated sperm that swim to the egg produced by the female gametophyte, located within an ovule.

### *Gnetophytes*

**Gnetophytes** are represented by three living genera and 95 species of plants that are very diverse in appearance. In all gnetophytes, xylem is structured similarly, none have archegonia, and their strobili (cones) have a similar construction. The reproductive structures of some gnetophyte species produce nectar, and insects play a role in the pollination of these species.

*Gnetum,* which occurs in the tropics, consists of trees or climbing vines with broad, leathery leaves arranged in pairs. *Ephedra,* occurring only in southwestern North America and Southeast Asia, is a shrub with small, scalelike leaves (Fig. 23.19*c*). Ephedrine, a medicine with serious side effects, is extracted from *Ephedra. Welwitschia,* living in the deserts of southwestern Africa, has only two enormous, straplike leaves that fray as it gets older (Fig. 23.19*d*).

## Angiosperms

**Angiosperms** (Gk. *angion,* "vessel"; *sperma,* "seed") are the flowering plants. They are an exceptionally large and successful group, with roughly 370,000 species thought to exist—six times the number of all other plant groups combined. Angiosperms live in all sorts of habitats, from fresh water to desert and from the frigid north to the torrid tropics. They range in size from the tiny, almost microscopic duckweed to *Eucalyptus* trees over 100 m tall.

It would be impossible to exaggerate the importance of angiosperms in our everyday lives. Angiosperms include all the hardwood trees of temperate deciduous forests and all the broadleaved evergreen trees of tropical forests. Also, all herbaceous (nonwoody